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Documentation

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To everyone who has been a part of this project, directly or indirectly, thank you for being an integral part of this milestone achievement.

Purpose of documentation

This document aims to describe the key elements of the project, starting with the project idea, the software development life cycle of the project, the design of the user interface, and the technical aspects of the project in detail, including what parts will be implemented and which parts are to be considered future work. With all of this information, this document becomes an absolute reference for the designers and developers during the implementation phase of this project.

Abstract

Quadriplegia is a type of paralysis that affect both the arms and legs and lose normal motor function. significantly impacting an individual's ability to perform daily tasks and communicate effectively. It is caused by an injury to the spinal cord in the neck region or by certain medical conditions.

Our graduation project aims to address the challenges faced by quadriplegic individuals by developing an innovative assistive technology solution that offers a range of benefits using only their eye movements.

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CHAPTER 1

Introduction

1.1 Introduction:

Quadriplegia, a condition that results in the loss of motor function in all four limbs, presents significant challenges for individuals affected by it. This physical disability not only affects their ability to move but also impacts various aspects of their daily lives. People with quadriplegia often face considerable difficulties in performing routine tasks that others may take for granted. In terms of basic human rights, individuals with quadriplegia, like all people, should

have the right to equal opportunities, access to education, employment, and healthcare.

Our project introduces a specialized eye-tracking device for individuals with quadriplegia, enabling computer control through eye movements. Recognizing the challenges faced by those with quadriplegia in traditional computer access, we provide an alternative solution to enhance their overall quality of life.

1.2 Problem Statement:

Imagine waking up every day to a world where the simplest actions, like reaching for a cup or taking a step are distant memories. This is the daily reality for those with quadriplegia. Communication becomes difficult because there are not enough tools that everyone can use, and job opportunities are limited because of unfair treatment and a job market that isn't always welcoming. Fundamental rights like equal access to education and healthcare are often

denied. Despite these challenges, there's hope for a world where awareness grows, technology evolves, and societies become more inclusive.

1.3 Solution Overview:

Our project aims to develop a special technology for people with quadriplegia, with a primary focus on an eye-movement-controlled interface. Using advanced computer vision and AI technology to track and interpret real-time eye movements, transforming them into actionable commands for seamless computer interaction.

The system includes a camera and specialized software designed to adapt to individual preferences, providing users with the ability to perform various tasks solely through their eye movements.

Transforming their lives in several key areas:

- It facilitates communication through text input and voice synthesis.
- enhances environmental control by integrating with smart home systems.
- opens educational opportunities and online learning.
- enables entertainment and gaming.
- provides access to employment through remote work and productivity tools.
- fosters social interaction via social media and video calls.
- supports creative expression in art and writing.
- contributes to health monitoring.
- improve the overall quality of life for quadriplegic individuals.

1.4 Impact on Business:

Our project has a huge impact on businesses. With an increasing awareness of accessibility needs and technological advancements, there is a growing market

demand for solutions that empower individuals with severe mobility limitations to interact with computers effectively. Plus, our cost-effective and high-quality solution can improve productivity and user satisfaction, leading to overall business success.

1.5 Ways of Marketing:

Our eye-tracking device offers independence to those with quadriplegia, but effective marketing is essential for its success. This guide covers diverse strategies, including digital advertising and partnerships, to promote the device. Our aim is to empower individuals with disabilities and foster inclusivity through technology.

- **Targeted Online Advertising**

  Digital Platforms: Utilize online advertising platforms such as Google Ads, Facebook Ads, and LinkedIn Ads to reach individuals with quadriplegia, caregivers, and healthcare professionals.

- **Social Media Marketing**

  Engagement and Awareness: Maintain active profiles on social media platforms such as Facebook, Twitter, and Instagram to engage with your target audience, share success stories, provide product updates, and raise awareness about the benefits of your eye-tracking device.

  Community Building: Join relevant online communities and forums where individuals with disabilities and caregivers gather to share experiences, ask questions, and seek support. Participate in these communities authentically, offering valuable insights and assistance.

CHAPTER 2

Related Work

2.1 Related Work:

Past research has explored devices using eye-tracking technology to aid people with physical disabilities. These studies help us understand how eye control can be effectively implemented.

some works aimed at achieving the purpose of our project have been carried out, but in different ways and techniques. Some of these works are devices with eye-tracking

technology or assistive technologies for individuals with quadriplegia include:

- **Tobii Dynavox EyeMobile Plus:**

The EyeMobile Plus is the first purpose built, consumer-style bracket solution for mobile computer access and communication. By combining eye tracking, speech recognition, switch and IR in one product, it allows for completely hands-free access to Windows 10 tablets. This gives individuals with physical and communication impairments the ability to navigate, control and access apps, internet, music, e-books, social media, games and more using the natural movements of the eyes.

Tobii Dynavox EyeMobile Plus

- **SmartNav:**

SmartNav is an assistive technology device that uses infrared cameras to track head movements, allowing users to control the computer cursor by moving their head. It's designed to assist individuals with various mobility challenges.

The device tracks the movement of a reflective dot placed on the user's forehead or other part of the body, translating these movements into mouse cursor movements on the computer screen. This technology enables individuals with limited mobility, such as those with spinal cord injuries or neuromuscular disorders, to navigate and interact with computer interfaces more independently.

SmartNav

- **Eyegaze Edge:**

A special eye tracking camera mounted below the Eyegaze Edge screen observes one of the user's eyes. Sophisticated image processing software analyzes the camera's images 60 times each second and determines where the user is looking on the screen. A user operates the system by looking at rectangular "keys" or cells that are displayed on the control screen. To "press" a key, the user looks at the key for a specified period of time. The gaze duration required to visually activate a key, typically about ½ second, can be adjusted by the user. An array of menu keys and exit keys allow the user to navigate the Eyegaze programs independently.

Eyegaze Edge

- **Quadstick:**

The QuadStick is a mouth operated game controller for Quadriplegics. The QuadStick is available in three models, the original QuadStick, the Singleton, and the QuadStick FPS. The primary difference between the two game controller models is the joystick module. The QuadStick FPS has a larger, more precise joystick than the original model.

QuadSticks game controllers have four sip & puff sensors and a lip position sensor, all connected to a 32-bit ARM processor that converts the sensor inputs into USB and Bluetooth signals for PC's & Game consoles.

The connections between the input sensors and the signals sent to the host are configurable by the user and can be quickly changed between pre-configured profiles, while playing a game, to match different situations in the game. Specific configuration profiles can easily be created for games that require combinations or unique control inputs.

The Quadstick comes in three different models to fit your needs and budget.

The Magic Arm kit

The Articulated Arm kit

The Flexible Arm kit

The Quadstick needs a mounting arm to hold it in front of the user.
Quadstick offers three choices in mounting arm kits.

- **PCEye Explore by Tobii:**

PCEye is a compact eye tracker that enables people with physical disabilities to control a Windows computer with their eyes. Replace a traditional keyboard and mouse with your eyes and surf the web, connect on social media, play games, create documents and more. Simply connect the PCEye to your Windows device via USB, and access everything your computer has to offer, intuitively and independently with TD Control.

2.2 Project Challenges:

Developing a reliable and accurate eye-tracking system can be extremely challenging for us, Here are some of the challenges that our project may face:

- Our project must be both affordable and of high quality. This means we need to find a balance between keeping our costs low and ensuring that our work meets the highest standards.

- Ensuring precise tracking of eye movements and minimizing errors requires advanced computer vision algorithms and hardware integration.

- Users may require time to adapt to the new eye-controlled interface. Designing a user-friendly system is essential to facilitate a smooth transition and ensure user acceptance.

2.3 Features:

- **Eye-Tracking Technology:** advanced eye-tracking technology to accurately respond to users' eye movements in real-time.

- **Computer Control:** Users with quadriplegia can navigate computer interfaces, control applications, and interact with digital content solely through their eye gestures.

- **Healthcare Application:** In addition to computer control, our project includes a specialized healthcare and management application designed to cater to the specific needs of people with disabilities.

- **User-Friendly Design:** Our application is easy to use, with a simple interface and smooth design to make it accessible and comfortable for users.

- **Integration with Standard Interfaces:** Our project seamlessly integrates with standard computer interfaces, making it compatible with a wide range of devices and applications.

2.4 Risk Management:

First, we would like to explain the concept of risk; Risk management involves anticipating unexpected events that could impact your project, either positively or negatively. These events can affect various aspects of the project, including people, processes, technology, and resources. It's essential to differentiate between risks and issues, risks are

potential future events, while issues are current problems that need to be addressed.

To manage risks effectively, it's crucial to understand your project objectives thoroughly. This allows you to identify potential obstacles that could hinder the team's ability to achieve desired outcomes. Connie Emerson emphasizes that risk management entails identifying threats to project objectives and addressing them proactively from the outset.

It's important to note that risks can be either positive or negative. While project managers often focus on threats, risks can also present opportunities for improvement or advancement.

Common types of project risks include:

- **Cost:** Events that could lead to budget overruns.
- **Schedule:** Unplanned conflicts or delays that affect project timelines.

- **Performance:** Events that result in outcomes not meeting project specifications.
- **Security:** Risks related to the loss of control over

Project Area	*Project Risk*	*Risk Type*	*Solution*
Project Scope Management	Unclear or evolve requirements in terms of expected deliverables	Project Scope	Write a CR document to outline expected Customer Requirements and SRS document to outline expected System Requirement.
Resource Management	Lack of rightly skilled or experienced resources	Skills	choose the skills you need to implement your project, the person who would be responsible for each part, and each person begins to develop his skills so that he can implement what is required of him.
Project security	the potential loss of control over personal information	security	Implement encryption, access controls, and regular security audits to safeguard personal information.
Technical Development	Technical complexities in eye-tracking technology	Technical	Conduct thorough testing, and collaborate with experts

personal information.

So now let’s look at some of our project risk:

CHAPTER 3

Project Methodology

3.1 Introduction:

This chapter will discuss the agile software engineering process used in this project. Agile development is a methodology focused on iterative and incremental progress, with requirements and solutions evolving through collaboration among self-organizing, cross-functional teams.

3.2 Software development life cycle (SDLC):

Before explaining our SDLC methodology, let's first understand what SDLC means. In essence, a software

development life cycle is a roadmap for working on a digital solution.

In other words, SDLC is a blueprint designed for a team to create, maintain, and fix digital products. The steps of a software development life cycle process depend on the project size and project goals.

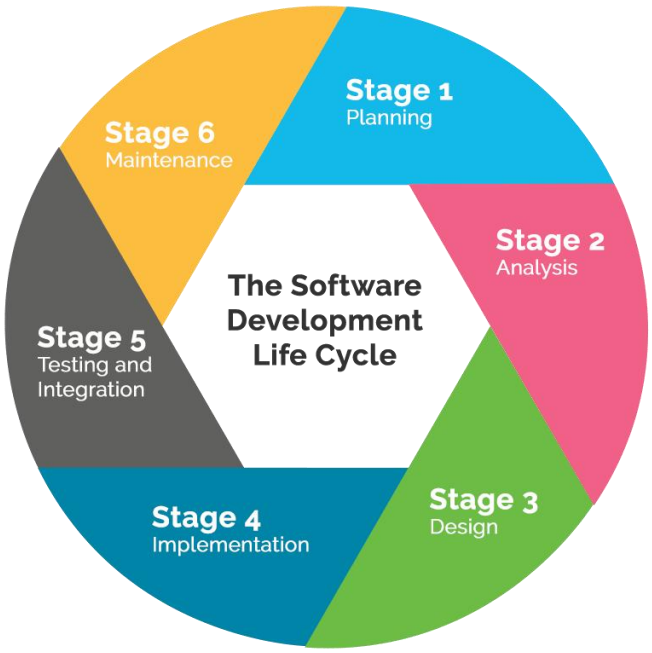
Each step also provides a general map of how it should be completed. In most cases, every development team creates it software development cycle or adopts one of the models, which we will explore further.

The following figure is a graphical representation of the various stages of a typical SDLC. A typical Software Development Life Cycle consists of the following stages, as shown in Figure 3.2

Let’s explain each phase in the SDLC what mean?

<u>Planning:</u>

The planning phase is the foundation of the SDLC. During this phase, the project's scope, objectives, and constraints are clearly defined. This



involves developing a comprehensive project plan that outlines tasks, timelines, and resources. The planning phase also includes risk assessment and management, where potential risks are identified, and strategies are devised to mitigate them. This phase sets the stage for the entire project by ensuring that everyone involved understands the goals and how to achieve them.

Analysis:

In this phase, the focus is on gathering detailed requirements from stakeholders, including customers, users, and other parties. These requirements are analyzed to ensure they are clear, complete, and feasible.

The output is a detailed requirement specification document that outlines both functional requirements (what the system should do) and non-functional requirements (performance, security, usability, etc.).

This document is essential for guiding the design and development phases and must be approved by stakeholders to ensure alignment with their needs and expectations.

Design:

The design phase translates the requirements into a blueprint for building the software.

This involves creating detailed architectural designs that define the overall structure of the system and how its components will interact.

Detailed design documents are also created, specifying the system's components, interfaces, data flow, and user interface design. These documents serve as a guide for developers and ensure that the system's design aligns with the requirements.

Implementation (Coding):

During the implementation phase, developers write the actual code to build the software. This phase involves following the design documents closely to ensure that the system is built correctly.

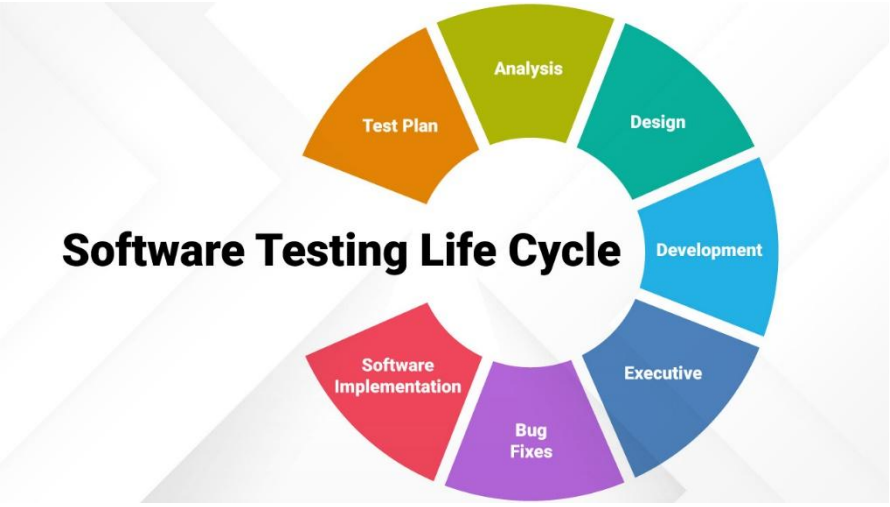
Developers adhere to coding standards and guidelines to maintain code quality and consistency. Unit tests are also developed and executed to test individual components for functionality and to catch any issues early in the development process.

Testing:

The testing phase involves rigorous testing of the software to ensure it meets the specified requirements and quality standards.

Various types of testing are performed, including unit testing (testing individual components), integration testing (testing the interaction between components), system testing (testing the entire system), and acceptance testing (validating the system with user requirements).

Any defects or bugs identified during this phase are documented and addressed. The goal is to ensure the software is reliable, functional, and ready for deployment.



Maintenance:

The maintenance phase begins once the software is deployed and in use. This phase involves monitoring the software for any issues, bugs, or performance problems that may arise.

Regular updates and enhancements are made to improve the software and address any newly discovered requirements or issues. Maintenance ensures the software remains effective, secure, and up-to-date over time, adapting to changing needs and environments.

3.3 Selecting Methodology:

Selecting a Software Development Life Cycle (SDLC) methodology can be challenging for many of us. This challenge often arises because few of us are familiar with the criteria for choosing a methodology that will add value to our project.

The best approach to selecting the right SDLC methodology is to carefully consider the project's needs and the development team's skills. There is no one-size-fits-all solution, and the most suitable methodology for a given project may vary depending on specific circumstances. Therefore, to select the appropriate SDLC methodology, it is essential to follow a structured process.

The process contains such as the following steps:

1. **Learn about the SDLC methodologies:**

To choose the right SDLC methodology, it's crucial to have sufficient experience and a thorough understanding of the various SDLCs available.

2. **Understanding the needs of Stakeholders:**

We must understand the business domain, stakeholder concerns and requirements, business priorities, our

technical capabilities, and technological limitations to select the appropriate SDLC based on these criteria.

3. **Select Criteria:**

In this step, you need to choose specific criteria or arguments to compare different methodologies and select the appropriate SDLC.

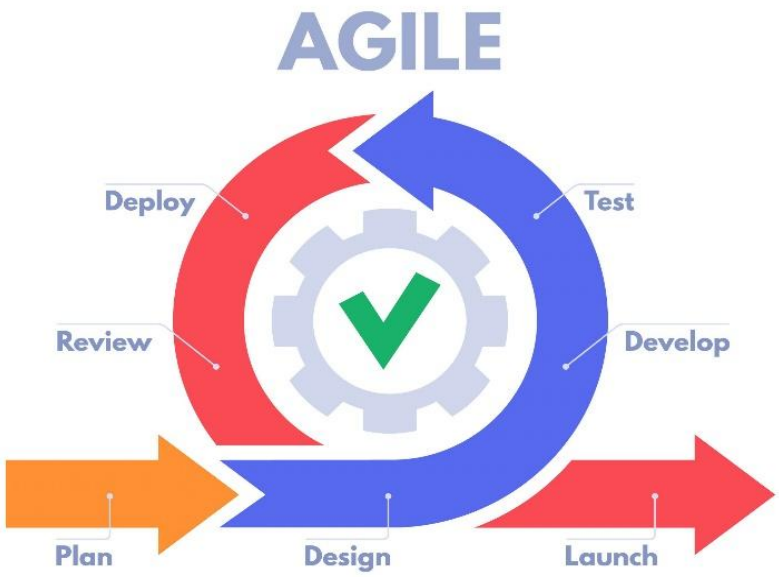
4. **Decide:**

In this step, after completing the comparison process, you engage in a discussion with your team to make your final decision.

- ✓ After following these steps, we decided to use **Agile Methodology**.

3.4 Agile Methodology:

Agile is a software development methodology that emphasizes flexibility, collaboration, and iterative progress. Unlike traditional approaches that follow a linear, step-by-step process, Agile allows for continuous improvement and adaptation throughout the development lifecycle.



❖ **Key characteristics of Agile include:**

- **Iterative Development:** Agile projects are divided into small, manageable units called iterations or sprints. Each iteration typically lasts from one to four weeks and results in a working product increment.

- **Collaboration:** Agile promotes frequent communication and collaboration among team members, stakeholders, and customers. This ensures that everyone is aligned on project goals and requirements.

- **Flexibility:** Agile embraces change and accommodates evolving requirements, allowing teams to respond quickly to feedback and market changes.

- **Customer Involvement:** Customers and end-users are actively involved throughout the development process, providing feedback at regular intervals to ensure the product meets their needs.

- **Continuous Improvement:** Agile teams regularly reflect on their processes and outcomes to identify areas for improvement. This fosters a culture of continuous learning and adaptation.

- **Cross-functional Teams:** Agile teams are typically cross-functional, comprising members with diverse skills and expertise necessary to complete all aspects of software development within each iteration.

- **Adaptive Planning:** Rather than detailed upfront planning, Agile relies on adaptive planning where requirements and solutions evolve through collaboration and feedback.

Agile methodologies such as **Scrum**, **Kanban**, and **Extreme Programming (XP)** are popular implementations of Agile principles, each with its own practices and frameworks tailored to different project needs and team dynamics.

Agile has gained widespread adoption due to its ability to deliver value quickly, improve team productivity, and enhance customer satisfaction through frequent and incremental releases of working software.

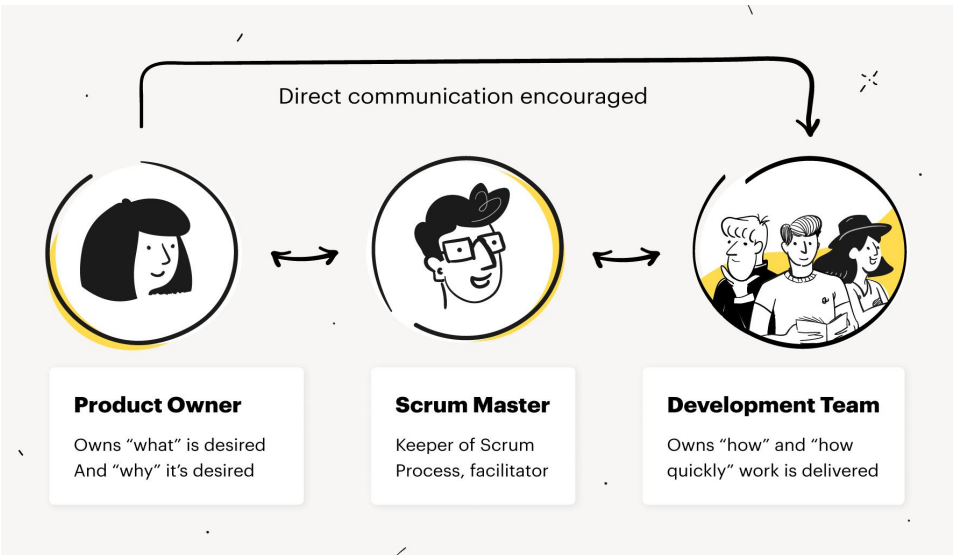
3.5 Scrum

Scrum is a popular Agile framework for managing and executing software development projects. It is characterized by its iterative approach and emphasis on collaboration, flexibility, and continuous improvement.

Here are the key components and principles of Scrum:

1. Roles:

- **Product Owner:** Represents the stakeholders and manages the product backlog, prioritizing features based on business value.
- **Scrum Master:** Facilitates the Scrum process, removes obstacles, and ensures the team adheres to Scrum practices.
- **Development Team:** Cross-functional group responsible for delivering increments of potentially shippable product at the end of each sprint.



2. Main Artifacts:

- **Product Backlog:** A prioritized list of features, enhancements, and fixes that represent the requirements for the product. Managed and prioritized by the Product Owner.

  Sprint Backlog: The set of items selected from the Product Backlog for a specific sprint, along with a plan for delivering the product increment and achieving the sprint goal.

- **Product Increment:** The sum of all the completed product backlog items at the end of a sprint. It must be in a usable condition and potentially releasable.

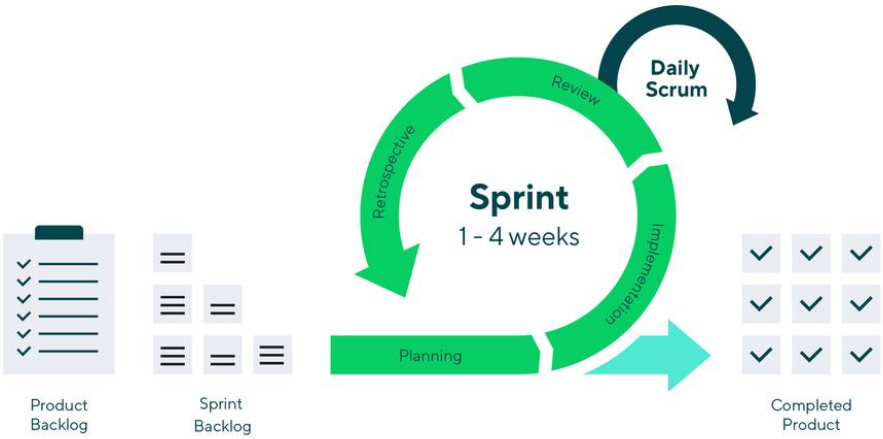
3.6 Scrum Life Cycle

- **Sprint:** A time-boxed period (usually 1-4 weeks) during which a potentially releasable product increment is created.

- **Sprint Planning:** A meeting where the Scrum Team collaborates to select items from the Product Backlog for the upcoming Sprint and creates a plan for achieving the Sprint Goal.

- **Daily Scrum (Daily Standup):** A short (15-minute) daily meeting for the Development Team to synchronize activities, discuss progress, and identify any impediments.

- **Sprint Retrospective:** A meeting where the Scrum Team reflects on the Sprint and identifies improvements for the next Sprint.



3.7 Advantages of Scrum

- **Flexibility and Adaptability:** Scrum's iterative approach allows teams to respond quickly to changes in requirements or priorities. This flexibility enables teams to deliver valuable increments of work more frequently, increasing adaptability to market dynamics and customer feedback.

- **Improved Transparency:** Scrum promotes transparency through its defined roles, artifacts, and events. This transparency ensures that everyone involved

in the project has a clear understanding of goals, progress, and challenges, fostering better communication and alignment.

- **Enhanced Customer Satisfaction:** Scrum involves customers and stakeholders throughout the development process, gathering feedback during sprint reviews and adjusting priorities based on evolving needs. This continuous engagement helps ensure that the delivered product meets or exceeds expectations.

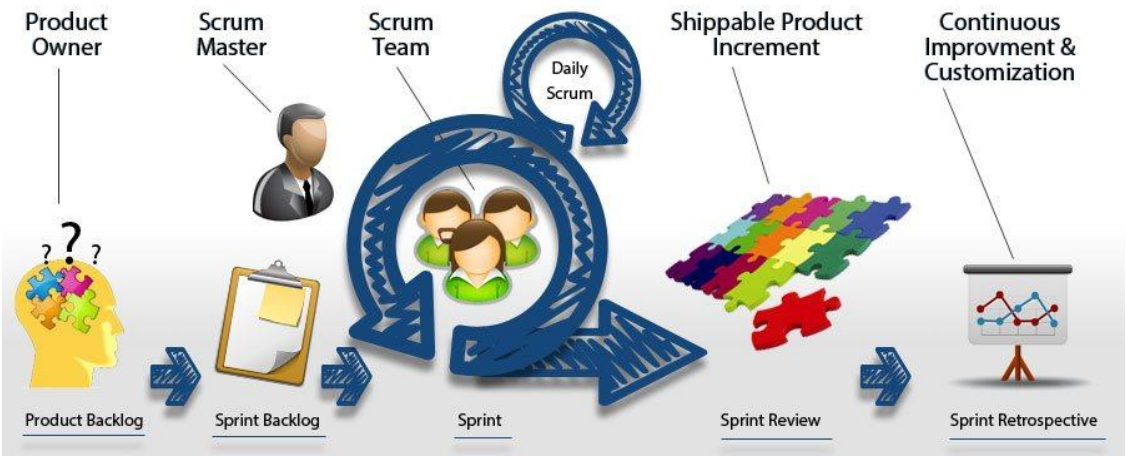
- **Higher Product Quality:** By focusing on delivering a potentially shippable product increment at the end of each sprint, Scrum encourages rigorous testing, collaboration among team members, and adherence to quality standards. This results in higher-quality products and reduced defects.

- **Increased Productivity:** Scrum emphasizes self-organizing teams and empowers them to make decisions related to how they will accomplish their work. This autonomy, combined with clear goals set during sprint planning, motivates team members and fosters productivity.

- **Better Risk Management:** Scrum's iterative and incremental approach enables early identification and mitigation of risks. Issues and challenges are addressed early in the process, minimizing their impact on project timelines and outcomes.

- **Continuous Improvement:** The sprint retrospective allows teams to reflect on their performance, identify areas for improvement, and implement changes in subsequent sprints. This focus on continuous improvement fosters a culture of learning and innovation within the team.

- **Predictable Delivery Schedule:** Scrum's time-boxed sprints and the regular cadence of events (sprint planning, daily scrum, sprint review, sprint retrospective) provide a predictable rhythm to the project. This predictability helps stakeholders and customers plan accordingly and manage expectations.



3.8 Disadvantages of Scrum

- **Complexity in Implementation:** Scrum's framework and principles may be challenging for teams new to Agile practices to implement effectively. It requires a shift in mindset and often involves significant changes in team dynamics and organizational culture.

- **Lack of Prescriptive Guidance:** Scrum provides a lightweight framework with defined roles, events, and artifacts, but it does not prescribe specific engineering practices or tools. This flexibility can lead to inconsistencies in how Scrum is applied across different teams or projects.

- **Time and Resource Intensive:** Scrum ceremonies, such as sprint planning, daily scrums, sprint reviews, and retrospectives, require dedicated time and commitment from team members. This can be resource-intensive, particularly for teams with competing priorities or limited availability.

- **Dependency on Team Collaboration:** Scrum relies heavily on collaboration and communication among team members, stakeholders, and customers. If there are

communication breakdowns or conflicts within the team, it can impact productivity and hinder project progress.

- **Handling of Changing Requirements:** While Scrum is designed to accommodate changes in requirements through iterative development, frequent changes or unclear priorities can lead to scope creep or difficulty in maintaining focus on sprint goals.

- **Suitability for Large or Complex Projects:** Scrum may face challenges in scaling to larger teams or projects with extensive dependencies and integration points. Additional frameworks or scaling techniques (e.g., Scrum of Scrums, LeSS, SAFe) may be required to manage complexity effectively.

3.9 Conclusion

In conclusion, Agile Scrum is a highly favored project management framework among software development teams. Its iterative and incremental approach facilitates frequent software releases and adaptable responses to changes. Moreover, emphasizing collaboration and communication enhances team morale and productivity significantly.

CHAPTER 4

Care Mobile Application

4.1 Introduction:

Our application aims primarily to provide care for patients who are unable to move by having them permanently connected to companions, in addition to providing many services related to the patient's medical appointments and data within the application.

4.2 Target Audience:

- Primary Users

Patients: Especially those with chronic conditions or limited mobility.
Elderly: Individuals who may need assistance managing their health.

- Secondary Users
  Caregivers: Family members or professional caregivers who assist patients.

4.3 The User Interface:

First, we need to know that the user interface is designed based on two important concepts UX and UXD.

user experience (UX)

User experience, or UX, is a term used to describe the overall experience a user has when interacting with a product or service in a given context. Depending on how the product or service is designed, the experience can range from delightful to downright frustrating! You'll often hear about UX in relation to digital products, such as websites and apps— but UX isn't limited to the digital space. Anything that can be experienced can be designed, from the packaging of a toothbrush to the wheels of an orthopedic chair. The impact of good (and bad!) UX is everywhere. That's one of the reasons it's such an exciting field, and also explains why you already know a lot more about UX than you realize. Every time you curse a push door that has a pull bar, or close a confusing website in frustration, you're making a judgment on the quality of its UX design.

So, UX is all about the user's interaction or experience with a product or service.

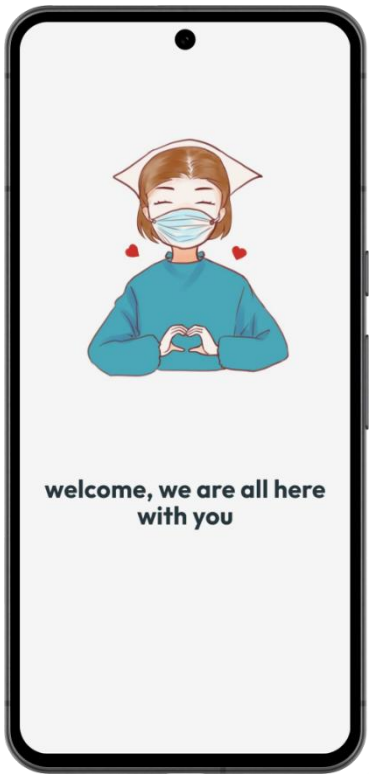
user experience design (UXD)

User experience design, or UXD, considers each element that shapes the user experience. It's all about designing specifically for the needs of the user or customer, looking at things like ease of use, quality, and efficiency. If you think about it, humans can only benefit from complex systems like websites and computer programs (or any product, really) if these systems are somewhat user-friendly.

4.4 Application's Screens:

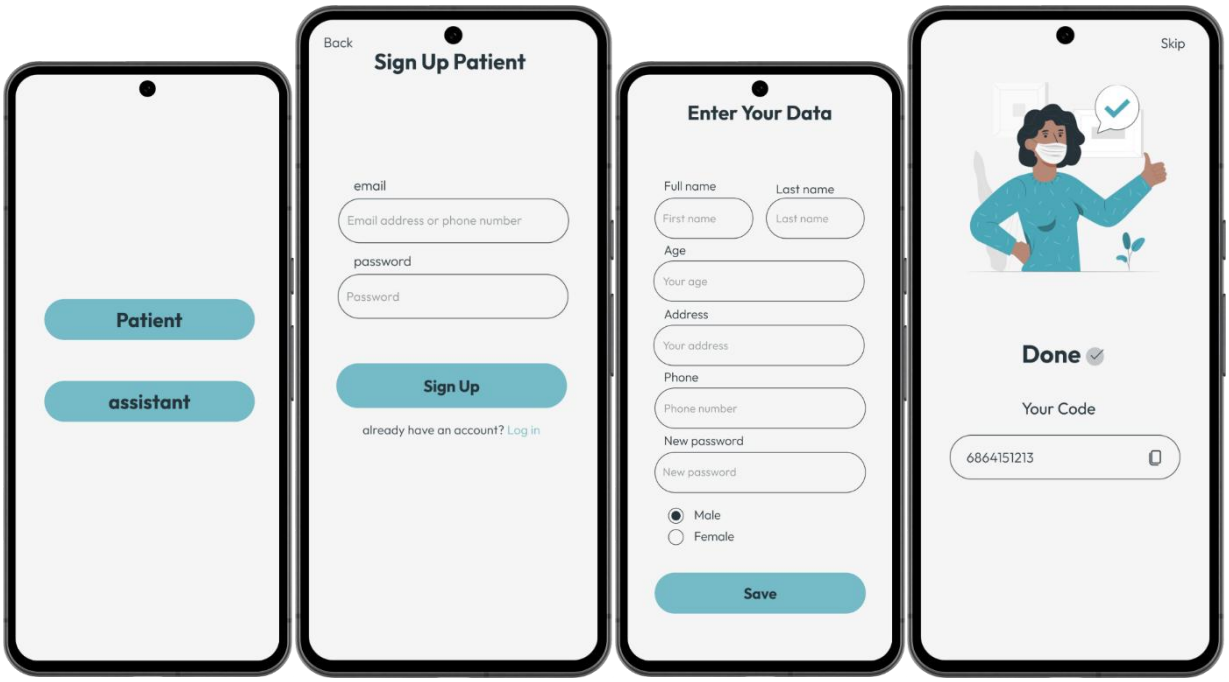
Splash Screen

This opens the screen of the application when it appears for a few seconds, then we move to the next screen of logging in



Sign up as a patient

In the beginning, the user chooses whether he wants to sign up as a patient or as an assistant to the patient. When he selects the patient, the sign up page will appear to him. If the patient has registered before, he will go to the login page and enter the application via email and password. If he has not registered before, it will appear to him. Sign up pages, as shown in the pictures. He creates an account, then enters his data, and after confirming these steps, each patient appears with his own code, through which the patient and the assistant are linked.

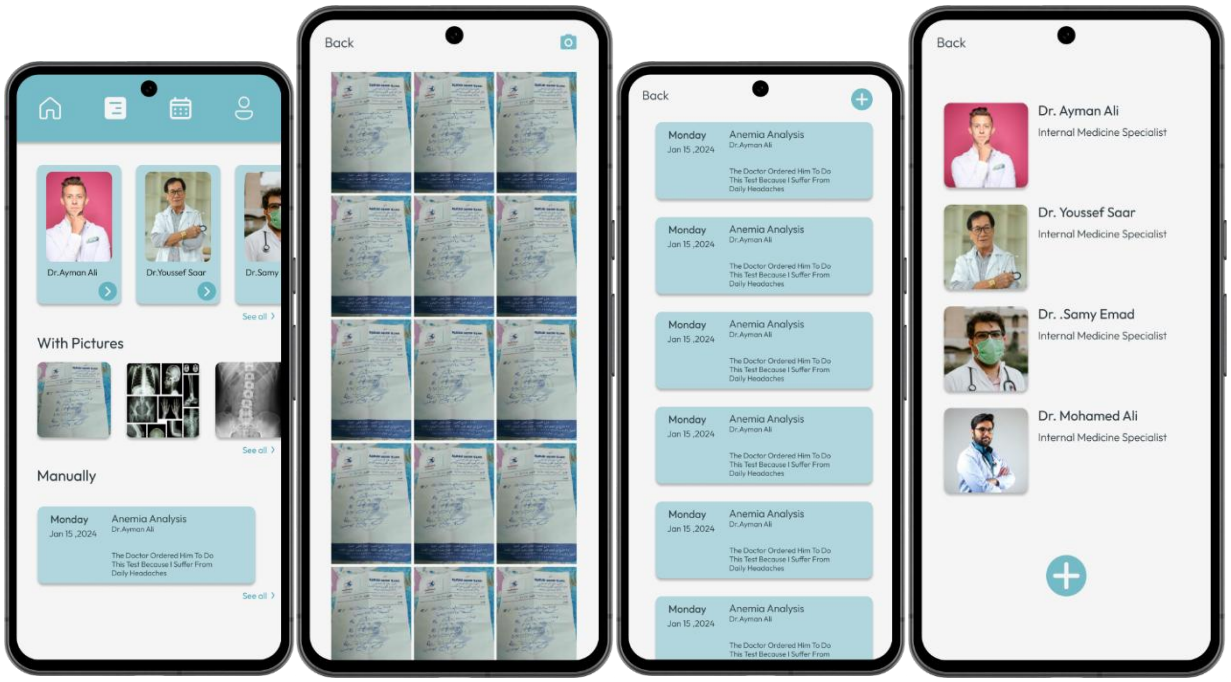


Patient home pages

The main page for the patient contains a tasks section, and the next task appears as the background for this section. It also contains a section for all the assistants associated with this patient, and each one of them can be contacted individually, in addition to the fact that at the bottom of the page there is a button to contact all of them in the event of an emergency.

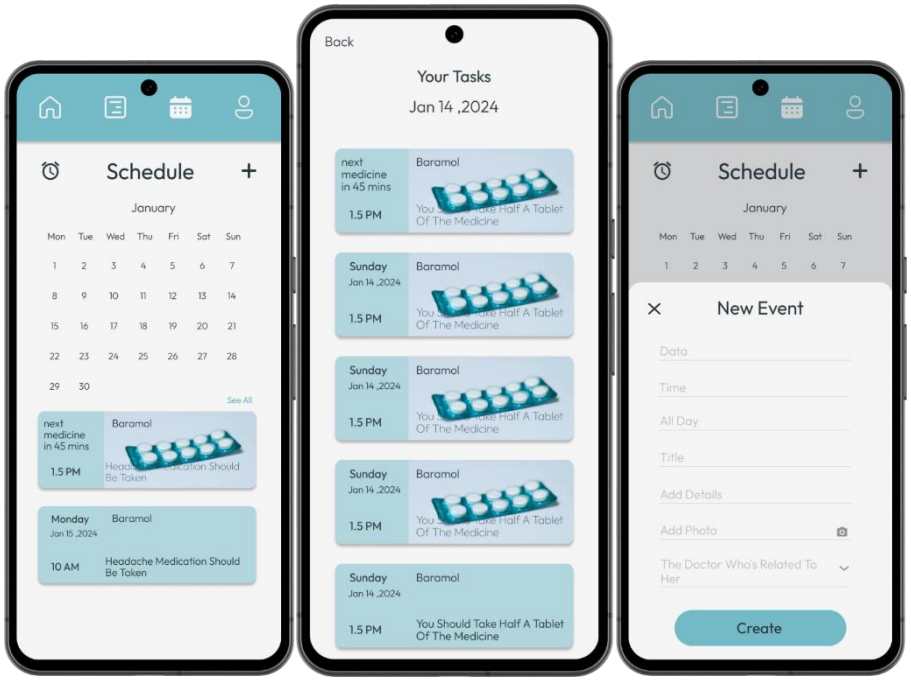
Patient data pages

The patient's data page contains three sections for the doctors treating him and the data that needs to be stored in the form of images and data entered manually.



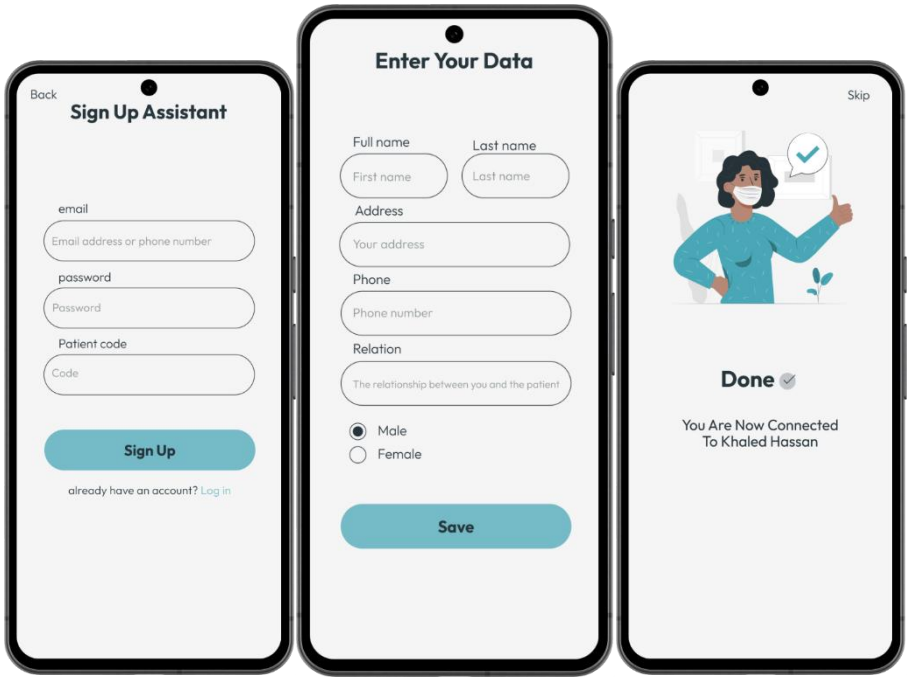
Patient schedule pages

The patient's appointment schedules page contains an alarm to remind the patient of the tasks required of him, and a calendar. When you click on any date on it, the tasks required for this day appear, in addition to a button that allows you to add a new task.



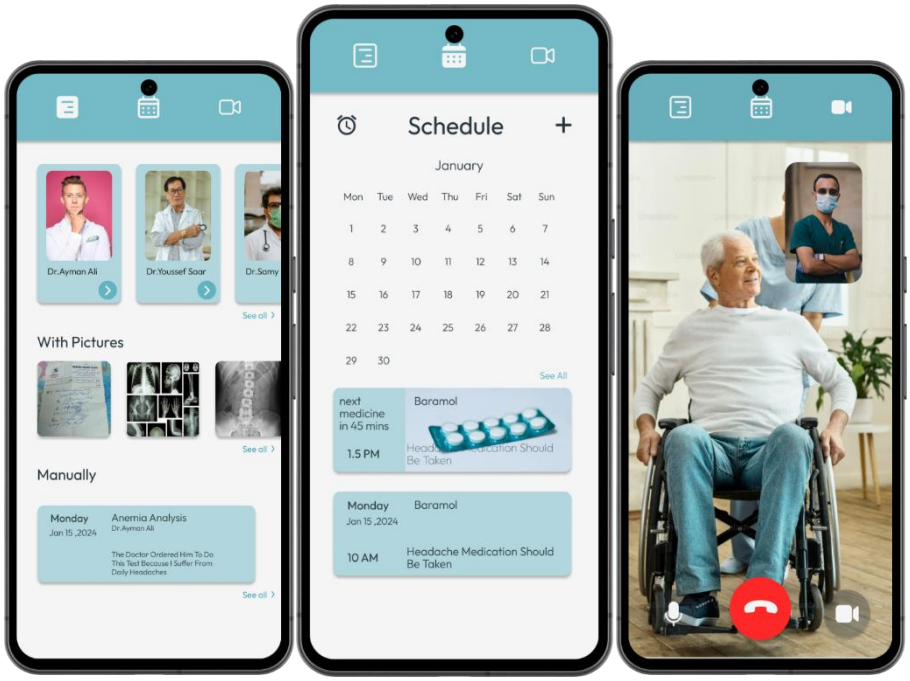
‘Sign up as an Assistant

The assistant signs up to the application using the code that was given to the patient during his sign up. After completing the sign up, the assistant's data is linked directly to the patient who owns that code.

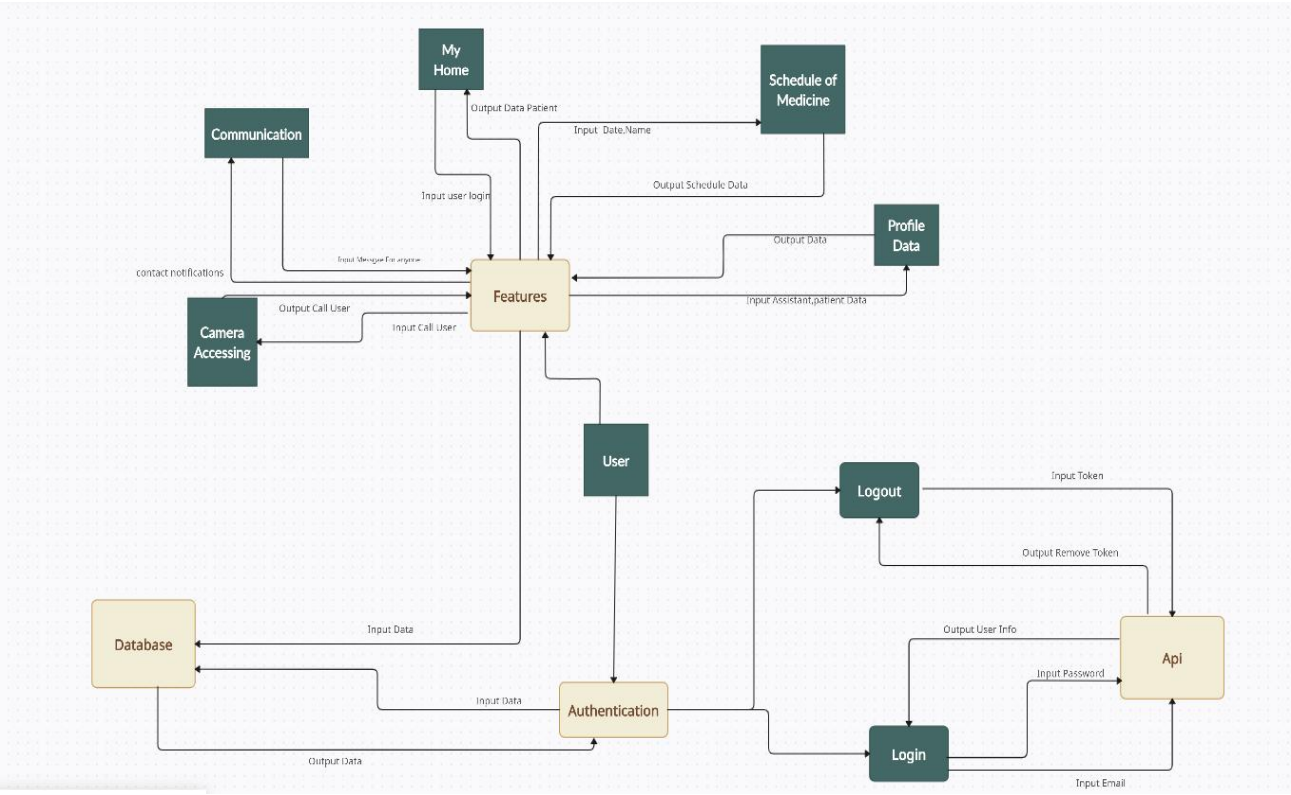


Pages that the assistant can access

The assistant can access and modify the patient’s data, and he can also access the patient’s tasks schedule, and he can also video call the patient



4.5 Software Context:



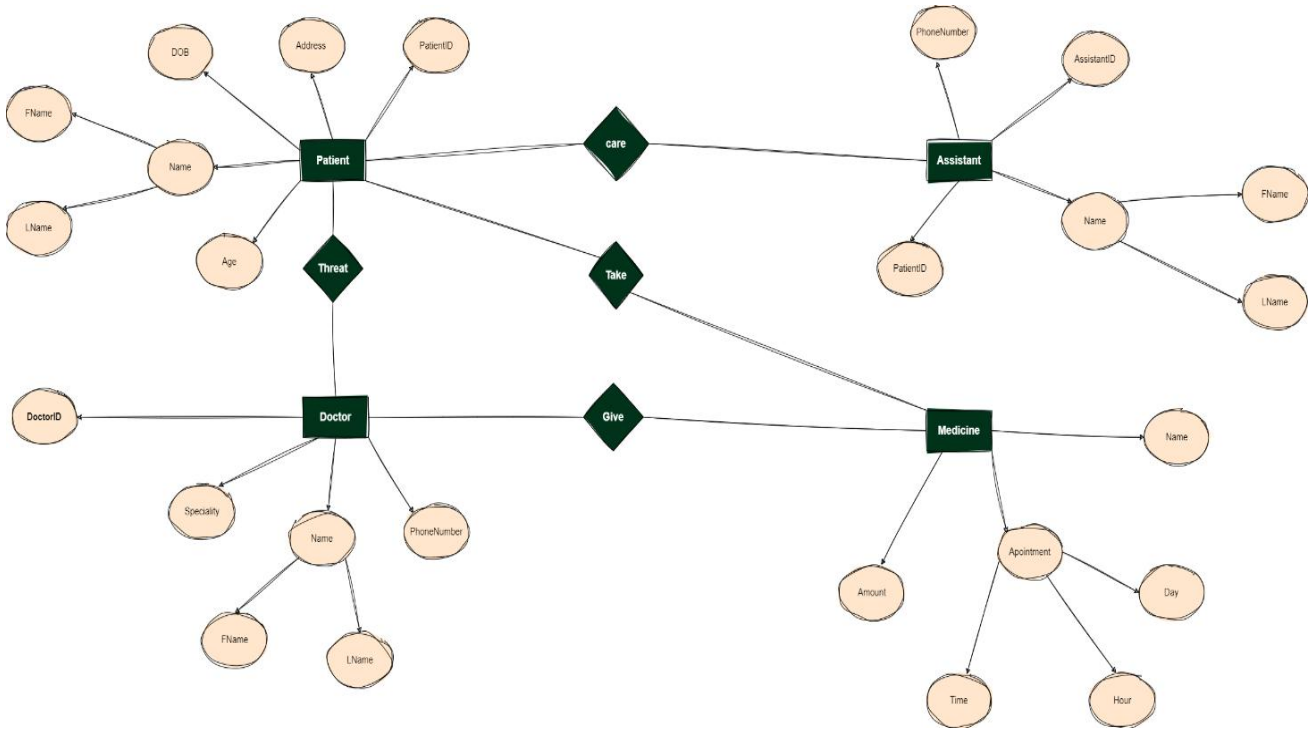
4.6 System Requirements Specification:

Req_ID	**Req_001**
Description	User shall login to a pre-existing account.

Input	Email and Password	***Output***	
Req_ID	**Req_002**		
Description	Patient shall validate entered email and p assword.		
Input	Email and Password	***Output***	
Req_ID	**Req_003**		
Description	Checking for email and password exist and each email matching his password.		
Input	Email and Password	***Output***	
Req_ID	**Req_004**		
Description	Home Screen shall show feature in app		
Input		***Output***	
Req_ID	**Req_005**		
Description	Patient can call Assistant video call		
Input	Patient Call	***Output***	
Req_ID	**Req_006**		
Description	All information about doctors		
Input	Name,speciaity,address	***Output***	
Req_ID	**Req_007**		
Description	Patient can show Rays with picture		
Input	Picture ray,details,who doctor	***Output***	
Req_ID	**Req_008**		
Description	Patient can show Rays manually		
Input	Name,Who Doctor,Time	***Output***	

Req_ID	**Req_009**		
Description	Medicine shall linked Schedule to remember		
Input	Date,Time,Photo,Name	***Output***	

Req_ID	**Req_0010**		
Description	Patient can show and edit medicine		
Input	Name,uses,who doctor,photo	***Output***	Save medicine info
Req_ID	**Req_0011**		
Description	Patient shall edit his profile through settings		
Input		***Output***	Edit his profile
Req_ID	**Req_0012**		
Description	User can edit or add assistant through request		
Input		***Output***	Confirm or remove
Req_ID	**Req_0013**		
Description	Users shall specifies his Language		
Input		***Output***	English or Arabic
Req_ID	**Req_0014**		
Description	Users shall send Rate App		
Input		***Output***	Rate



4.7 ERD Diagram:

CHAPTER 5

Project Software tools and Technologies

5.1 Introduction:

This chapter describes the software tools and technologies used in the development of Eye Tracker for people with quadriplegia project. The goal of this chapter is to provide a comprehensive overview of the tools and technologies used, as well as their role in the project.

The software tools and technologies used in the development of project include:

- Applications, systems and frameworks
- Platforms and environments
- API's
- Languages

- Architecture dependencies

5.2 Applications and frameworks:

Mobile

- **Figma:** It is a powerful design tool primarily used for UI/UX design and prototyping. It allows multiple users to collaborate in real-time, making it an excellent choice for teams working on digital projects.

Here are some key features and functionalities of Figma:

Real-Time Collaboration: Multiple designers can work on the same project simultaneously, see each other's changes in real-time, and leave comments directly on the design.

Prototyping: Figma provides tools to create interactive prototypes, allowing designers to simulate user interactions and transitions between screens. This helps in testing and refining user flows before development.

Design Systems: Figma supports the creation and management of design systems, including reusable components, styles, and assets. This ensures consistency across different projects and speeds up the design process.

Vector Networks: Figma's unique approach to vector editing allows for more flexibility and control compared to traditional vector tools. You can create and manipulate shapes and paths with ease.

Plugins and Integrations: Figma has a wide range of plugins and integrations that extend its functionality. These include tools for animation, accessibility, version control, and more.

Cross-Platform: Figma is web-based, meaning it can be used on any operating system with a web browser. There's also a desktop app available for macOS and Windows.

Version History: Figma automatically saves and versions your design files, allowing you to revert to previous versions or review the history of changes.

Hand-off to Developers: Figma makes it easy to share designs with developers, providing detailed specifications, CSS, and code snippets for implementation.

- **Android Studio:** It is the official integrated development environment (IDE) for Google's Android operating system. It is built on JetBrains' IntelliJ IDEA software and designed specifically for Android development.

Android Studio offers several features that make it a powerful tool for developing Android apps, **including**:

- A code editor that provides syntax highlighting, code completion, and linting
- A debugger that allows you to step through your code and inspect its
- execution.
- A profiler that can help you identify performance bottlenecks in your app.

- An emulator that allows you to test your app on a variety of Android devices.
- A range of tools for managing your Android project
- Android Studio is free to download and use. It is available for Windows, macOS, and Linux.

Android Studio is an integrated development environment (IDE) optimized for Android app development. It's free to download and use.

Here are some key features of Android Studio:

- **Code Editor and Error Analysis**:
  - Android Studio provides a powerful code editor with syntax highlighting autocompletion, and error analysis.
  - You can navigate to type definitions and search for type usages.
  - It supports error analysis and warnings to improve code quality.

- **Visual Editor and User Interface Preview**:
  - You can design and preview your app's user interface using the Layout Editor in Android Studio.
  - It helps you visually arrange UI elements.

- **Code Debugger and Execution**:
  - Android Studio allows you to run your app on an emulator or a real device.
  - It supports debugging and hot reload for quick updating changes.
- **Full IDE Integration**:
  - You can develop both Android and Flutter apps within the same environment.
  - It provides full support for Dart development, code analysis, and hot reload.

- **Active Community and Rich Resources**:

- Flutter has a large community of developers and abundant online resources.
- You can find tutorials, articles, examples, and useful libraries.

- **Visual Studio Code:** is a free, open-source code editor developed by Microsoft. It's designed for building and debugging modern web and cloud applications

Here's an overview of its key features and functionalities:

- VS Code is known for its performance and speed. It’s lightweight, which makes it a popular choice for developers who need a fast and efficient coding environment.
- It works on Windows, macOS, and Linux, allowing developers to use the same editor across different operating systems.
- It provides intelligent code completions based on variable types, function definitions, and imported

modules. It enhances productivity by predicting and suggesting code as you type.

- It includes a powerful debugger that allows developers to run and debug their code directly within the editor.
- The editor has a built-in terminal, which lets developers execute shell commands without leaving the coding environment.
- VS Code supports code snippets for quick and efficient coding. It also offers various formatting options to keep the code clean and consistent
- VS Code provides shortcuts for writing HTML and CSS code.

VS Code's combination of performance, flexibility, and rich feature set makes it a favorite among developers for a wide range of programming tasks.

- **MongoDB:** is a widely used, open-source NoSQL database management system designed for handling large volumes of data and providing high performance,

availability, and scalability. It is known for its document-oriented data model and flexible schema.

Here are some key features and concepts related to MongoDB:

- **Document-Oriented Storage:** MongoDB stores data in flexible, JSON-like documents called BSON (Binary JSON). Documents are grouped into collections, analogous to tables in relational databases.
- **Schema Flexibility:** MongoDB does not require a predefined schema, allowing documents within a collection to have different structures. This makes it easy to evolve the data model without downtime.
- **High Performance:** MongoDB is designed to handle many read and write operations per second, making it suitable for high-traffic applications.

- **Indexing:** MongoDB supports various types of indexes (single field, compound, geospatial, text, etc.) to improve query performance.
- **Aggregation Framework:** MongoDB includes a powerful aggregation framework that allows for complex data processing and analysis operations, such as filtering, grouping, and transforming data.
- **Full-Text Search:** It has built-in support for full-text search, allowing users to perform text searches on indexed fields.
- **ACID Transactions:** multi-document ACID transactions are supported, allowing for more complex operations with guaranteed consistency.

Here are some of the benefits of using MongoDB:

- MongoDB is a popular choice for web and mobile applications.

- The document-oriented nature of MongoDB makes it ideal for managing content that varies in structure.
- MongoDB’s powerful aggregation framework and scalability make it suitable for big data and analytics applications.
- MongoDB can handle the massive volumes of data generated by IoT devices efficiently.

MongoDB's combination of performance, flexibility, and ease of use makes it a popular choice for modern application development.

- **Postman**: Postman is an API Platform for developers to design, build, test and iterate their APIs. we made the access endpoints to the database through the user interface, we have to make sure that they are correct, if the data is sent or received correctly, to ensure the validity and integrity of these end points.

Benefits of Using Postman:

- **Efficiency:** Simplifies API development and testing processes, saving time and effort.
- **Collaboration:** Facilitates teamwork through shared collections, environments, and documentation.
- **Versatility:** Handles a wide range of HTTP requests and responses, making it suitable for various API testing scenarios.
- **Learning Tool:** Useful for learning and experimenting with APIs, especially for beginners.

- **GitHub**: It is a platform and cloud-based service for software development and version control using Git, allowing developers to store and manage their code. It provides the distributed version control of Git plus access control, bug tracking, software feature requests, task management, continuous integration, and wikis for every project. One of the methods of successful communication in the world of software between developers and some of

them is their full understanding of the electronic clouds. We communicated on the most famous work platforms, uploaded the project, distributed the chats between the team members, and uploaded the work to the main project on the work platform so that everyone in the teams can see what happened, so he can send a comment on the work. Or share it or download it for modification We used GitHub to work as a cohesive team, so each of us can see what happened in terms of development, addition or modification in any part of the project

5.3 Platforms and environments:

API'S

- **REST**: is a web service that conforms to the principles of REST (Representational State Transfer) architecture. It

allows different software applications to communicate over the web using standard HTTP methods and can be used to perform CRUD (Create, Read, Update, Delete) operations on resources

Key Concepts of REST APIs:

- **Resources:** resources are the fundamental units of data. Each resource is identified by a unique URL (Uniform Resource Locator).
- **HTTP Methods:** REST APIs use standard HTTP methods to perform operations on resources (GET, POST, PUT, DELETE, PATCH)
- **Client-Server Architecture:** The client and server are separate entities that interact with each other. The client makes requests to the server, and the server processes these requests and returns appropriate responses.
- **Uniform Interface:** It includes the use of standard HTTP methods, URIs for resource identification, and

standard media types like JSON or XML for resource representation.

Benefits of REST APIs:

- **Simplicity:** REST APIs use standard HTTP methods and status codes, making them easy to understand and use.
- **Scalability:** Statelessness and the ability to cache responses enhance scalability.
- **Flexibility:** REST APIs can handle a wide range of data formats and can be used with any programming language.

REST APIs are a popular and powerful way to create web services that are scalable, maintainable, and easy to use.

5.4 Languages:

Mobile

- **JavaScript:** It is a high-level, dynamic programming language commonly used in web development to create interactive and dynamic web pages. It is one of the core technologies of the World Wide Web, alongside HTML and CSS, and is supported by all modern web browsers without the need for plugins

Key Features of JavaScript：

- **Client-Side Scripting:** JavaScript runs in the user's browser, allowing for interactive features such as form validation and animations without needing to communicate with the server.
- **Server-Side Scripting:** With environments like Node.js, JavaScript can also be used on the server side to handle requests, manage databases, and perform other back-end tasks.
- **Interpreted Language:** JavaScript code is executed line-by-line by the browser's JavaScript engine, without the need for prior compilation.

- **Event-Driven:** JavaScript supports event-driven programming, which allows code to be executed in response to user actions
- **First-Class Functions:** Functions in JavaScript are first-class objects, meaning they can be stored in variables, passed as arguments to other functions, and returned from other functions.
- **Asynchronous Programming:** JavaScript provides mechanisms for asynchronous programming, such as callbacks, promises, and async/await, which are essential for handling operations like network requests without blocking the main thread.

JavaScript's versatility and extensive ecosystem make it a powerful language for both client-side and server-side development. It continues to evolve, introducing new features and capabilities to meet the demands of modern web and application development.

- **Dart**: It is a modern but already mature programming language designed to make developers happier. It's concise, safe, interoperable with Java and other languages, and provides many ways to reuse code between multiple platforms for productive programming. It is a powerful language that is well-suited for a wide variety of tasks. It is especially popular for Android development, but it can also be used for web development, server-side development, and more.

Dart is syntactically like Java. But has some key differences, including:

- **Null safety:** Dart is a null-safe language, which means that you do not have to worry about null pointer exceptions.
- **Extensions:** Dart allows you to extend existing classes with new functionality.

- **Data classes:** Dart has a special type of class called a data class, which is
- **designed to be used for storing data.**
- **Lambdas:** Dart has a concise syntax for writing lambda expressions.

Here are some additional benefits of using Dart:

- **It is concise:** Dart's concise syntax makes it easier to write and read code.
- **It is safe:** Dart's null safety eliminates the risk of null pointer exceptions.
- **It is interoperable:** Dart can be used with Java, which makes it easy to migrate existing Java code to Dart.
- **It is expressive:** Dart's syntax allows you to express your ideas concisely

Machine learning

5.5 Architecture Dependencies:

Android

BLoC: It is which stands for Business Logic Component, is a popular design pattern and architectural concept extensively employed for managing state in Flutter apps.

- **Separation of Concerns:**
  - BLoC helps separate the UI-related code (widgets) from the business logic.
  - It ensures that your app's logic is organized and maintainable.
- **State Management:**
  - BLoC manages the app's state by handling events and emitting new states.
  - It provides a predictable way to update the UI based on changes in data or user interactions.

- **Event-Driven Architecture:**
  - BLoC receives events (user actions, data updates, etc.) and processes them.
  - It transforms events into new states, which are then reflected in the UI.
- **Testing and Reusability:**
  - BLoC makes it easier to write unit tests for your app's logic.
  - You can reuse BLoCs across different parts of your app.

Dio: It is an HTTP client library for Dart that simplifies making API requests in Flutter applications. It provides an easy-to-use API for handling HTTP requests and managing responses.

Here are some key points about Dio:

- **Making HTTP Requests:**

Dio allows you to make various types of requests, such as GET, POST, PUT, and DELETE.

- **Interceptors:**

  Dio provides interceptors to modify requests and responses. You can add custom headers, handle errors, or log network traffic using interceptors.

- **Cancellation:**

  Dio supports cancellation tokens to cancel requests that are no longer needed.

SharedPreferences: It is a crucial tool in Flutter for locally storing data. It allows you to persist information even after the user closes the app, ensuring that data remains accessible across sessions.

Here are the key points about SharedPreferences:

- **What is SharedPreferences?**

SharedPreferences is a simple key-value storage mechanism. It's used to store small amounts of data (e.g., user preferences, settings, tokens) on the user's device. Data stored in SharedPreferences remains available even after the app restarts.

- **How Does It Work?**

SharedPreferences stores data as key-value pairs. You can save and retrieve values using unique keys. It's lightweight and efficient for storing primitive data types (bool, double, int, String, and List<String>).

- **Usage in Flutter:**
  - To use SharedPreferences in Flutter, add the shared_preferences package to your project.
  - The plugin wraps SharedPreferences on Android and NSUserDefaults on iOS.

- You can save, retrieve, and delete data easily.

Intl: It is library plays a crucial role in internationalization and localization. Let's dive into what it does:

- **What is intl?**

The intl library provides tools for handling internationalization-related tasks. It covers various aspects, including:

- **Message formatting and replacement:** For localized strings.
- **Date and number formatting and parsing:** Ensures consistency across different locales.
- **Bidirectional text support:** Handles right-to-left (RTL) languages.

- **How Does It Work?**

You can create an Intl instance for a specific locale. It allows you to format dates, numbers, and messages according to that locale.

The library ensures that your app adapts to different languages and cultural conventions.

Machine learning

CHAPTER 6

Embedded System

6.1 Introduction:

An embedded system is a specialized computing system that performs dedicated functions or tasks within a larger system. Unlike general-purpose computers, which can run a wide variety of applications, embedded systems are designed to perform specific tasks, often with real-time computing constraints. They are typically embedded as part of a larger device and are usually optimized for efficiency, reliability, and performance.

6.2 Project Definition:

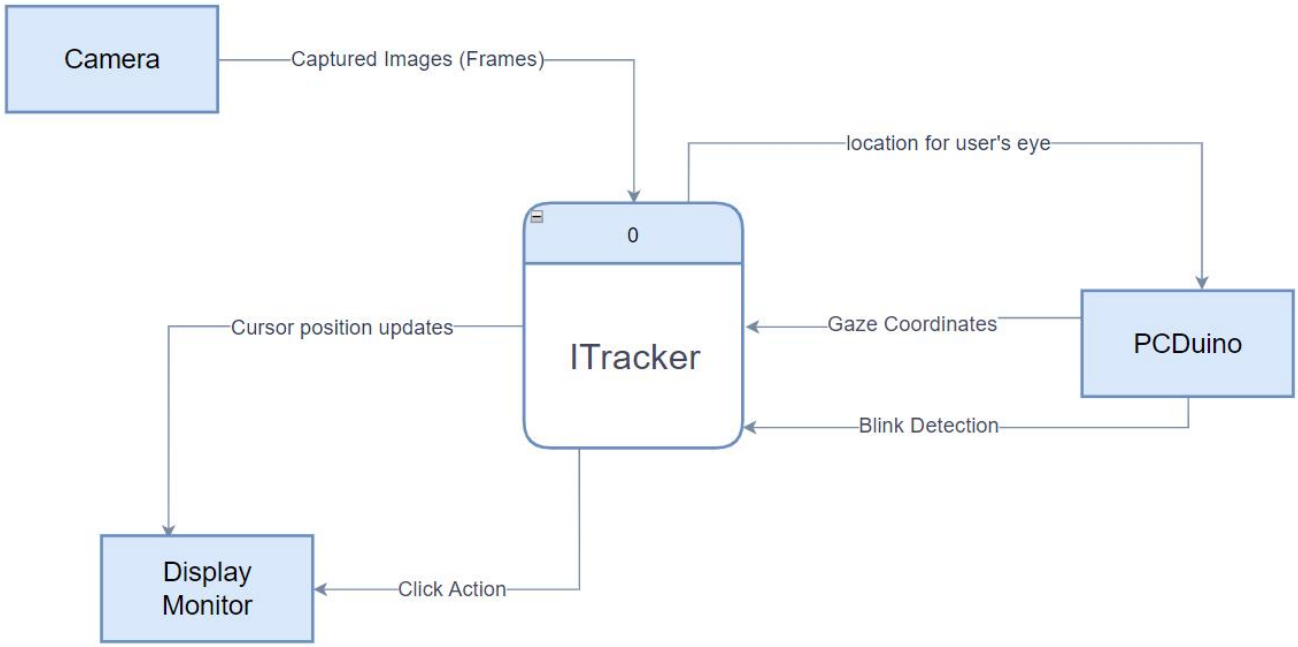
Our project involves developing an eye-tracking mouse that uses the PCDuino board and its integrated camera to move the cursor based on the user's eye movements. The embedded system within this project handles image capture,

processing, gaze estimation, and cursor control, providing a seamless and intuitive user experience.

6.3 Features:

- The Eyeme can be calibrated for different users.
- The Eyeme can control the cursor based on eye movements.
- The Eyeme can perform click actions using blink detection.
- The Eyeme provides visual indicators for cursor position and actions.
- The Eyeme allows customization of cursor speed and sensitivity.
- The Eyeme is compatible with Windows, macOS, Linux, Android, and IOS.
- The Eyeme supports hands-free operation, ideal for users with physical disabilities.
- The Eyeme compensates for slight head movements.

6.4 Software Context:



6.5 Software Requirements Specification:

N	*Requirements*	*Related Requirements*	*Related Stakeholders*	*Type*
1	Real-Time Eye Tracking	• Capture images from the camera at a minimum of 30 frames per second. • Preprocess images to enhance quality for gaze estimation.	Developers, Users	Functional
2	Gaze Estimation	• Use an AI model to estimate the user's gaze direction	Developers, Users	Functional

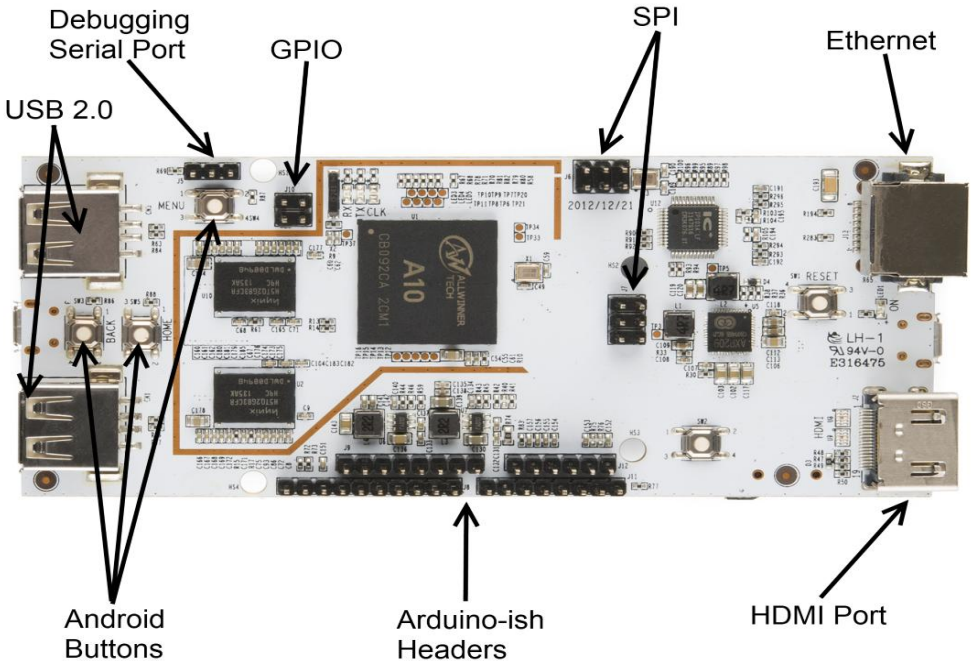
		• Convert gaze direction into screen coordinates		
3	Cursor Control	• Move the cursor based on the estimated gaze coordinates	Developers, Users	Functional
		• Provide smooth and responsive cursor movements		
4	Blink Detection	• Detect blinks to perform click actions	Developers, Users	Functional
		• Differentiate between intentional blinks for clicks and natural blinks.		
5	PCDuino	• Connect to the camera.	Developers	Functional
		• Integrate with model.		
6	Camera Module	• Detect user eyes.	Developers	Functional
		• Take the gaze coordinates.		
		• Send to the PCDuino		
7	Manage the integration process	• Ensure the computer vision code works correctly.	Developers	Functional
		• Ensure the PCDuino works correctly.		
		• Integrate between the PCDuino and the CV code		
8	Compatibility	• Compatible with Windows, macOS, and Linux operating systems.	Developers, Users	Non-Functional
9	Performance	• process and respond to eye movements with a latency of less than 50 milliseconds.	Developers, Users	Non-Functional

6.6 Hardware Components:

- **PCDuino Board**

The PCDuino board is chosen for its robust performance and compatibility with various sensors and peripherals. Key features of the PCDuino include:

- **Processor:** ARM Cortex-A8 1GHz
- **Memory:** 1GB DDR3 RAM
- **Storage:** Supports SD cards and onboard flash storage
- **Connectivity:** USB, HDMI, Ethernet, and GPIO interfaces



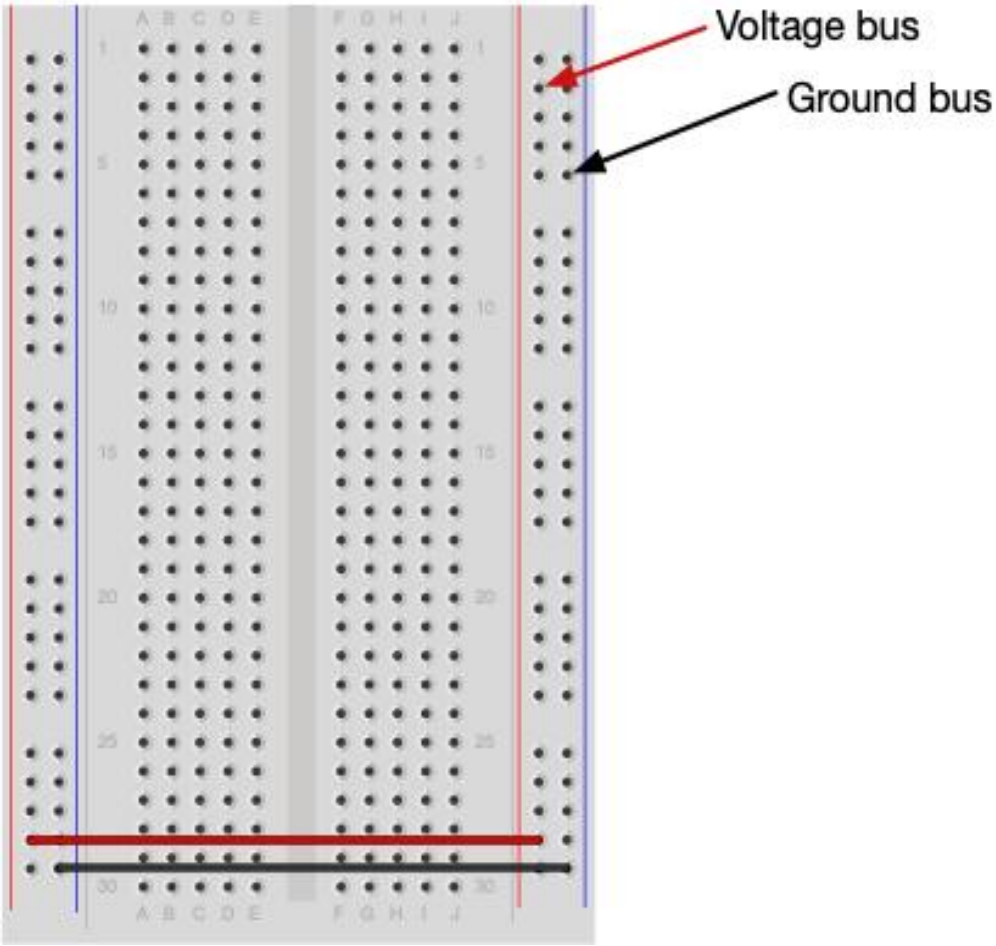
- **Camera Module OV7670**

This module is popular for its affordability and ease of integration with microcontrollers and single-board computers, making it suitable for various image capture and processing tasks in DIY projects and prototyping. Specifications include:

- **Resolution:** 640 x 480 pixels (VGA)
- **Frame Rate:** Up to 30 frames per second (fps) for VGA
- **Interface:** 8-bit parallel with SCCB (similar to I²C)
- **Features:** Automatic exposure, white balance, gain control, night mode, and frame rate control

- **Breadboard**

A breadboard is a reusable, plastic board with a grid of holes used for constructing and prototyping electronic circuits without soldering, making it easy to test and modify circuit designs.

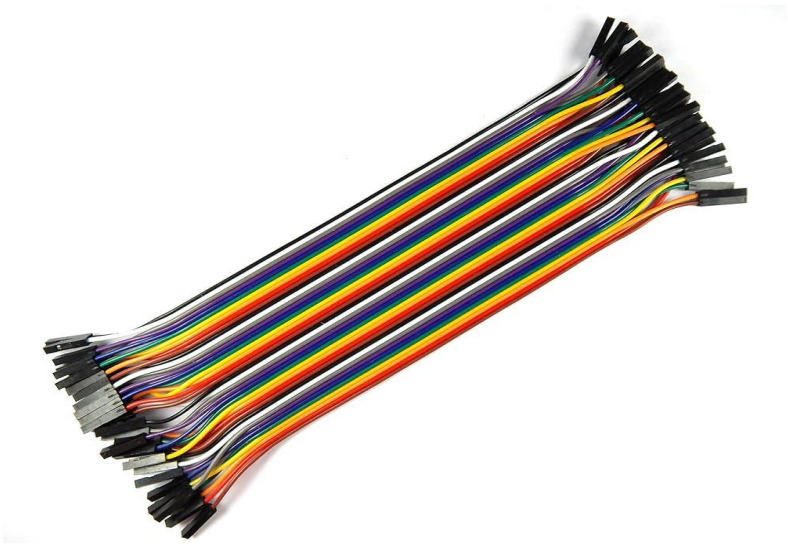


- **Jumper Wires**

Jumper wires are essential components in electronics used to make temporary connections between points in a circuit.

Types:

- **Male-to-Male:** Both ends have pins, commonly used to connect two points on a breadboard or between a breadboard and an Arduino.
- **Male-to-Female:** One end has a pin and the other a socket, used to connect a pin header to a breadboard or another component.
- **Female-to-Female:** Both ends have sockets, used to connect pin headers on different components.



6.7 Software:

- **Opearting System**

An operating system (OS) is software that manages computer hardware and provides services for computer programs. It acts as an intermediary between applications and the computer hardware, handling tasks such as memory management, file management, process management, and providing a user interface.

PCDuino OS:

The PcDuino runs on a Linux-based operating system, specifically Ubuntu, which is a popular choice for embedded systems due to its stability, extensive support, and compatibility with a wide range of hardware and software.

- **Software Components and Integration**

1. **Python Environment:** Ubuntu supports Python natively, which is essential for running our Python-based eye-tracking mouse application. The system can easily install necessary Python packages using pip or other package managers.
2. **Computer Vision Libraries:** Libraries such as OpenCV, Dlib, and PyTorch are well-supported on Ubuntu. These libraries are critical for tasks such as facial landmark detection, gaze estimation, and real-time video processing.
3. **Driver Support:** The Ubuntu operating system on PcDuino ensures that drivers for the built-in camera and other connected peripherals are readily available and easily installed.
4. **Development Tools:** Ubuntu provides a wide range of development tools, including GCC, GDB, and various

IDEs, which are beneficial for developing, debugging, and deploying our application.

- **Hardware Setup:**

1. Connect the power supply to the PCDuino board to provide stable power. Ensure the power supply meets the board's requirements (typically specified in volts and amps).
2. Connect the camera to one of the USB ports on the PCDuino board.
3. connect other USB peripherals such as keyboard, mouse, or additional storage devices as needed.
4. Connect the HDMI cable to its respective port on the PCDuino board for display output.
5. Once all connections are made, power on the PCDuino board using the connected power supply.

6.8 Environment setup for deploying machine learning models on PCDuino:

- **Software Setup and Integration:**

1. Install a suitable Linux distribution (e.g., Ubuntu, Debian) on the PCDuino board. Ensure it supports the board's hardware and required libraries for software development.
2. Ensure that necessary drivers for the webcam (camera) are installed and recognized by the operating system.
3. Install necessary libraries and packages such as:
    - OpenCV for image processing tasks, including capturing frames from the webcam.
    - PyTorch for running the AI model that interprets eye movements.
    - Dlib Library for machine learning, particularly used here for face detection and facial landmark detection.

- pynput for controlling the mouse cursor based on the AI model's output.

4. Upload the trained AI model to the PCDuino board.
5. Transfer the Python scripts that handle image acquisition, processing, and cursor control to the PCDuino board.
6. Configure these scripts to run automatically on startup to ensure continuous operation of the eye tracking system.

6.9 How to use Eyeme:

Follow these steps to set up and use Eyeme:

- First turn on the Eyeme device by connecting it to the power.
- Connect the Eyeme to your computer using a USB cable. Ensure the connection is secure and the device is powered on.
- Press the push button once to start the eye tracker mouse application.

- Once the application is running, the system will start tracking your eye movements.
- Move your eyes to direct the mouse cursor. The cursor will follow your gaze, allowing you to control it with your eye movements.
- Blink detection is enabled. A long blink will simulate a double mouse click. Ensure your blinks are deliberate to avoid accidental clicks.

✓ By following these instructions, you should be able to set up and use the eye tracker mouse effectively.

CHAPTER 7

Machine Learning

7.1 Introduction:

Eyeme deep learning models are used to apply an eye-movement-based mouse controller for disabled people, especially people with quadriplegia. This requires quick face detection and eye tracking using computer vision techniques.

7.2 Definition:

movement-controlled interface. Using advanced computer vision and AI techniques, the system simulates natural eye movement to allow these individuals to perform various tasks independently, enhancing their overall quality of life.

7.3 Features:

- **Face detection**
- **Eye detection**
- **Quick gaze estimation**

7.4 Key Elements:

- Facial Landmark Detection Model should:
    - Identifies and tracks 68 distinct facial landmarks.
    - Ensures robust facial feature detection for accurate gaze tracking.

- Operates in real-time to provide immediate feedback.

- Gaze Estimation Model (L2CSNet_Gaze360) should:
  - Estimates gaze direction in a full 360-degree range.
  - Utilizes deep learning for high accuracy in gaze estimation.
  - Minimizes errors to ensure reliable performance.

7.5 CUSTOMER REQUIREMENTS MAPPING

Req_ID	REQ_EYEME_ML_001_V1.0	mapping	The model should predict where the user is looking at the moment
Req_ID	REQ_EYEME_ML_002_V1.0	mapping	The model should be high accurate in natural time
Req_ID	REQ_EYEME_ML_003_V1.0	mapping	The model should work will at ant corner the user is looking from
Req_ID	REQ_EYEME_ML_004_V1.0	mapping	The model should work will even the light is not very good

7.6 Related work:

Gaze estimation, also called eye tracking, is a challenging field of research in computer vision. Researchers have tried to build a high accuracy model to track eye movement since around 2002 till now. There are two main methods to build such a model.

1. **Modelbased approaches**: modelbased approaches use a geometric model of an eye and can be subdivided into corneal-reflection-based and shapebased methods.
   - Corneal-reflection-based methods :

Those methods rely on external light sources to detect eye features.

- shape-based methods :

shape-based ones infer gaze direction from observed eye shapes, such as pupil centers and iris edges.

2. **Appearance-based methods**

directly use eyes as input and can potentially work on low-resolution images.
Appearance-based methods are believed to require larger amounts of user-specific training data as compared to model-based methods.

In this project we built a shape_based model but, due to hardware limitations, we finally used pre-trained models which were very useful.

7.7 The Custom Shape-Pased Model:

The model is designed using **TensorFlow** for eye tracking by identifying eye coordinates in facial images.

Methodology:

Dataset

The dataset consists of 1000 face images, which have been split into three sets: training, testing, and validation. Each image has been annotated to identify the eyes using the **labelme** library.

Data Preparation

- **Training Set**: Used to train the model.
- **Testing Set**: Used to evaluate the performance of the model.
- **Validation Set**: Used to fine-tune the model and prevent overfitting.

Data Loading and Preprocessing

- The images are read from the disk, resized to 250x250 pixels, and normalized by scaling the pixel values to the range [0, 1].
- Labels (eye coordinates) are read from JSON files and extracted.

Data Pipeline

The images and labels are combined into a single dataset. This dataset is then shuffled and batched to ensure efficient training. The same process is repeated for the testing and validation datasets.

Model Architecture

The model is constructed using a pre-trained ResNet152V2 as the base. Additional convolutional layers are added on top of the base model, followed by a reshaping layer to output the coordinates of the eyes.

Model Compilation

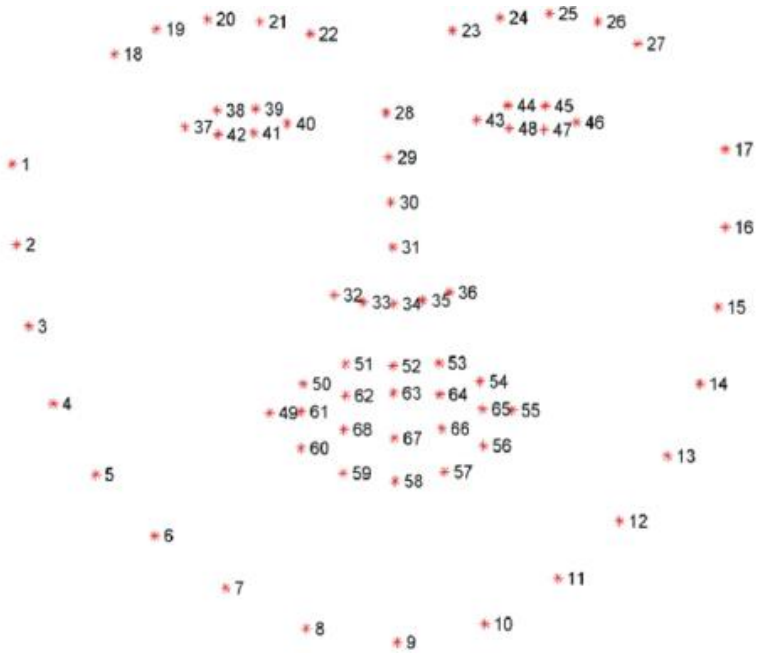
The model is compiled using the **Adam optimizer** and **Mean Squared Error** (MSE) loss function, which are suitable for **regression** tasks like predicting coordinates.

7.8 The eyeme model using pre-trained models:

7.8.1 68 Facial Landmarks

The 68 facial landmarks model is a commonly used facial landmark detection model in computer vision. It identifies 68 specific points on a human face, which are used for various applications such as facial recognition, emotion detection, and facial feature analysis.

is part of the Dlib library, which is a toolkit for making real-world machine learning and data analysis applications. The model is typically used for face alignment.



Eyes (Points 37-48):

- Points 37-42 represent the left eye.
- Points 43-48 represent the right eye.

Working Principle

1. **Face Detection**:
    - A face detector (like Dlib's HOG + SVM-based detector) locates faces in an image
2. **Landmark Localization:**

- The 68 landmarks model uses a **<u>shape predictor</u>** to identify and localize facial landmarks on the detected faces.

```
import dlib
import cv2
import numpy as np
```

Implementation:

```
detector = dlib.get_frontal_face_detector()
predictor = dlib.shape_predictor('shape_predictor_68_face_landmarks.dat')
```

Initialize Dlib's Face Detector and Shape Predictor

```
img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
faces = detector(gray)

```

Detect Faces in an Image

```
or face in faces:
    landmarks = predictor(gray, face)
    for n in range(0, 68):
        x = landmarks.part(n).x
        y = landmarks.part(n).y
        cv2.circle(img, (x, y), 2, (255, 0, 0), -1)
cv2.imshow('Landmarks', img)
cv2.waitKey(0)
```

Predict Landmarks for Each Detected Face:

Model Structure

1. Preprocessing:
    - The model typically starts with a face detection step to locate the bounding box of the face within an image. Dlib uses a HOG (Histogram of Oriented Gradients) feature descriptor combined with a linear SVM to detect faces.
2. Initialization:
    - An initial estimate of the landmark positions is made. This can be done using a mean shape (average of all landmark positions in the training dataset) aligned to the detected face bounding box.
3. Regression Trees:
    - The core of the model consists of an ensemble of regression trees. These trees are trained to predict the adjustments needed to improve the initial landmark estimates. The process involves multiple stages, where each stage refines the positions predicted by the previous stage.

Training Process

1. **Training Data:**

- The training dataset consists of facial images with annotated landmark positions.

2. **Feature Extraction**:
    - For each image, features around the ground truth landmark positions are extracted.
3. **Tree Training**:
    - Regression trees are trained in an ensemble. Each tree is trained to predict the offsets (adjustments) needed to move the initial or intermediate shape closer to the ground truth.
4. **Stage-wise Refinement**:
    - The training process involves multiple stages. At each stage, the trees are trained to correct the errors from the previous stage.

7.8.2 l2csNet_Gaze360

L2CSNet (**Look at Two Concurrent Stages Network**) is designed to estimate **where a person is looking** (gaze estimation) based on images or video frames of their face. The primary goal of gaze estimation is to **predict the gaze direction or the point where a person is looking**. L2CSNet approaches this task by processing images through two concurrent stages to enhance accuracy and robustness.

Key Components

1. **Architecture:**
    - **Two-Stage Network**: L2CSNet utilizes a two-stage network architecture where each stage processes the input image to refine the gaze estimation.
    - **Feature Extraction**: The first stage typically focuses on extracting features from the input image using convolutional neural networks (CNNs).
    - **Refinement Stage**: The second stage further refines these features to improve the accuracy of the gaze estimation.
2. **Convolutional Neural Networks (CNNs)**:
    - CNNs are employed to extract spatial features from the input images. These features are essential for understanding the facial landmarks and eye regions critical for gaze estimation.
3. **Loss Functions:**
    - L2CSNet often uses specialized loss functions designed to minimize the error in gaze direction prediction. This can include angular losses that measure the difference between predicted and actual gaze angles.

4. **Head Pose Estimation:**
    - The model may also incorporate head pose estimation as part of its pipeline, as head orientation significantly influences gaze direction.
5. **Training and Optimization:**
    - The model is trained on large-scale datasets with **annotated gaze directions**. Optimization techniques like **gradient descent** are used to minimize the loss functions.

Gaze360 Dataset

Gaze360 is a large-scale, 360-degree dataset used for training and evaluating gaze estimation models. It includes diverse head poses and gaze directions.

Key Characteristics

1. Data Collection:

- Gaze360 includes images and videos collected in various real-world environments, capturing people in different lighting conditions, backgrounds, and activities.

2. Annotations:

- Each image or video frame in the dataset is annotated with precise gaze directions and head

poses. These annotations are crucial for training gaze estimation models.

3. Diversity:

- The dataset includes a wide range of head poses, from frontal to extreme side views, and diverse gaze directions, making it challenging and comprehensive.

How L2CSNet Uses Gaze360

L2CSNet uses the Gaze360 dataset to train its two-stage network for improved gaze estimation. The diverse and comprehensive nature of Gaze360 allows L2CSNet to generalize well across different scenarios and head poses.

1. **Data Preprocessing**:
   - Images are preprocessed to normalize lighting conditions, align faces, and crop regions of interest focusing on the eyes and face.
2. **Feature Extraction and Refinement**:
   - The two-stage network processes these preprocessed images. The first stage extracts initial features, and the second stage refines these features to produce accurate gaze estimates.
3. **Evaluation:**

- The trained model is evaluated on a separate validation set from Gaze360 to ensure its accuracy and robustness.

Mean Error:

- 9.9 degrees on the Gaze360 dataset

7.9 CONCLUSION

Artificial intelligence (AI) is playing an increasingly important role in the development of Eyeme project. The models used in this project are assumed to help the user control his devices using his eyes only, putting in concern not to make a big stress on the eyes of the user.

As AI technology continues to evolve, we can expect to see even more sophisticated and powerful applications for eye tracking that would help people with disabilities to provide real-time communication with their environment.

We hope that our project will be a big step to encourage more people to search on this field.